

Exploratory Visualization of Fraud Risk Patterns in Bank Transaction Data

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1. Data Preparation

The full dataset was used for descriptive summaries and aggregated fraud-rate calculations. A small stratified sample was created only for code submission and sample-based plots. The dataset contains 1,000,000 transactions and 26 variables; `is_fraud` identifies fraudulent (1) and non-fraudulent (0) transactions.

Table 1: Dataset Dimensions

Measure	Value
Number of rows	1,000,000
Number of columns	26

Data-quality checks found no missing values, duplicate transaction IDs, duplicate full rows, or invalid values in the selected key variables, so no row removal or imputation was required before visualization.

Table 2: Overall Fraud Summary

Total transactions	Fraud count	Non-fraud count	Fraud rate
1,000,000	55,255	944,745	5.53%

There are 55,255 fraudulent and 944,745 non-fraudulent transactions, giving an overall fraud rate of 5.53%. Because this outcome is highly imbalanced, later comparisons use within-group fraud rates rather than raw fraud counts. Readable labels were created for fraud status, international status, night status, weekend status, and recent PIN change.

The prepared dataset retains all observations. A reproducible sample of 1,000 fraudulent and 1,000 non-fraudulent transactions was selected with seed 123 for compact code submission; full-dataset results are reported below.

2. Dataset Overview

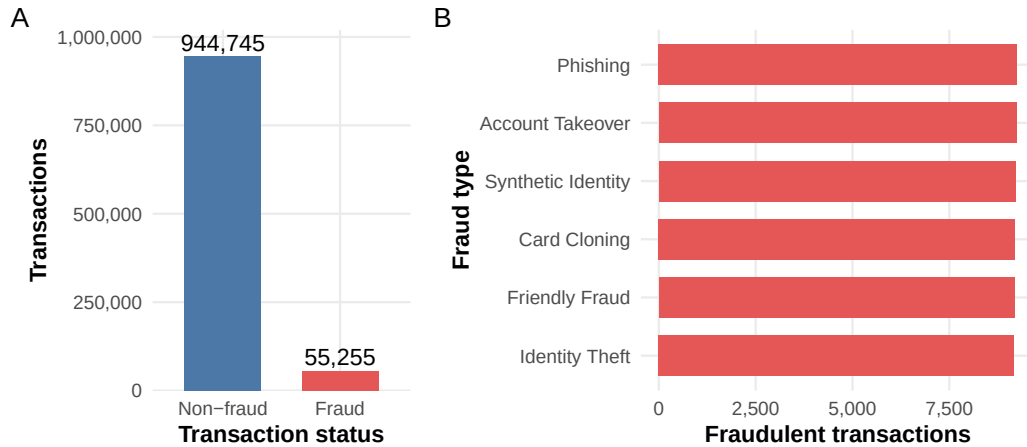


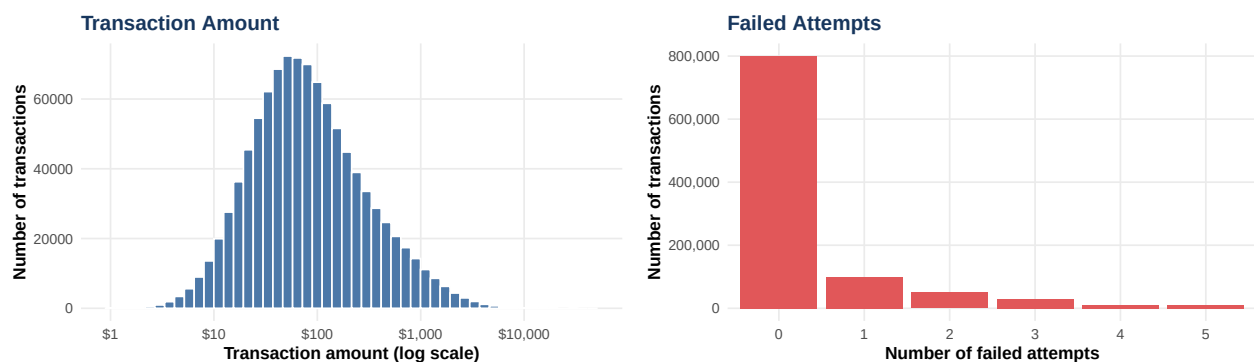
Figure 1: Compact overview of (A) outcome imbalance and (B) fraud-type composition.

Panel A shows that fraud is a minority outcome, representing 5.53% of all transactions; this imbalance motivates the use of rates in later comparisons. Panel B shows that the six fraud types are comparatively even, with no single type dominating the fraudulent subset. Together, the panels provide compact background context without competing visually with the risk-pattern results.

3. Univariate Analysis

Numerical variables

The following plots describe one variable at a time and do not compare fraud groups. Transaction amount uses a logarithmic x-axis because of its strong right skew.



Transaction amount is strongly right-skewed, which supports logarithmic scaling when comparing fraud groups. Failed attempts are concentrated at zero and one, with higher counts comparatively uncommon; this provides context for the later fraud-rate result. Univariate plots describe dataset composition but do not directly answer which characteristics are associated with fraud. Therefore, the bivariate and multivariate sections provide the main evidence for identifying fraud-risk patterns.

4. Bivariate Fraud Analysis

All comparisons below report the total number of transactions, fraud count, and fraud rate. The interpretation uses associative language and does not imply that any transaction characteristic causes fraud.

Binary transaction characteristics



Figure 2: Fraud rates across four binary transaction characteristics.

International transactions, night transactions, and transactions following a recent PIN change show visibly higher fraud rates than their comparison groups, while weekend and weekday rates are nearly identical. The first three patterns appear to be stronger visual indicators, whereas weekend status offers little visual separation. These associations directly address the research question but do not imply causation.

Merchant category

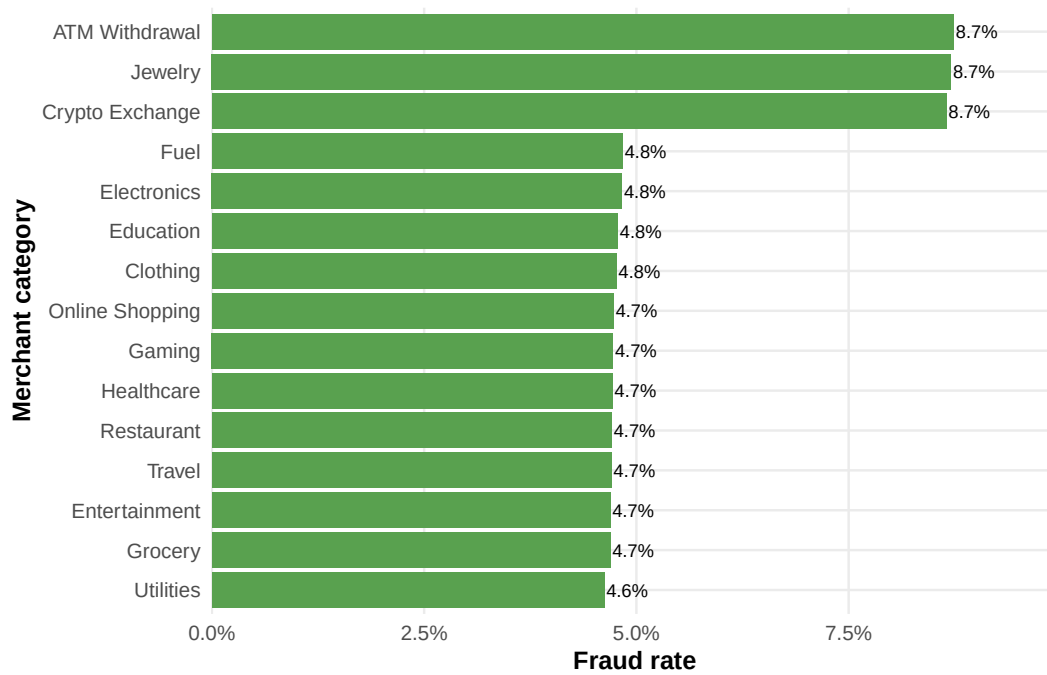


Figure 3: Within-category fraud rates and transaction counts by merchant context.

ATM withdrawal, jewelry, and crypto exchange transactions form a distinct upper group near 9%, while most other merchant categories cluster closer to the overall rate. Each merchant category contains approximately 66,000 transactions, so the rate differences are not driven by very small groups. Merchant context therefore appears strongly associated with fraud risk and is one of the stronger candidate figures.

Transaction amount by fraud status



Figure 4: Transaction amount by fraud status on a logarithmic scale.

The logarithmic boxplot shows substantial overlap between fraudulent and non-fraudulent transaction amounts. Transaction amount alone therefore provides weak visual separation between the two statuses. This cautionary result shows that an intuitively relevant variable need not be a strong standalone indicator.

Failed attempts

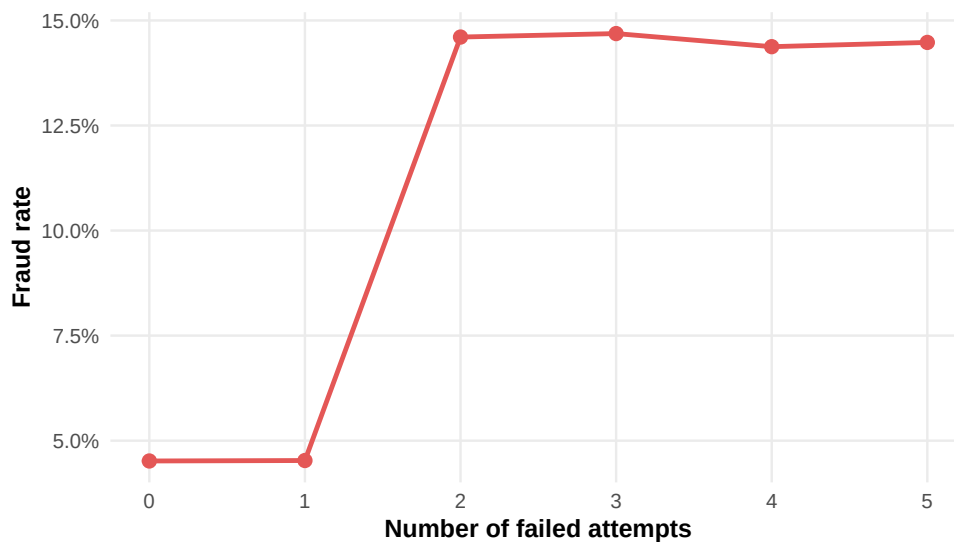


Figure 5: Fraud rate by number of failed attempts.

Fraud rates remain relatively low at zero or one failed attempt and rise sharply when the count reaches two or more. The ordered separation is stronger than in most other comparisons. Repeated failed attempts therefore appear strongly associated with elevated fraud risk, making this pattern useful for the figure shortlist.

Hour of day

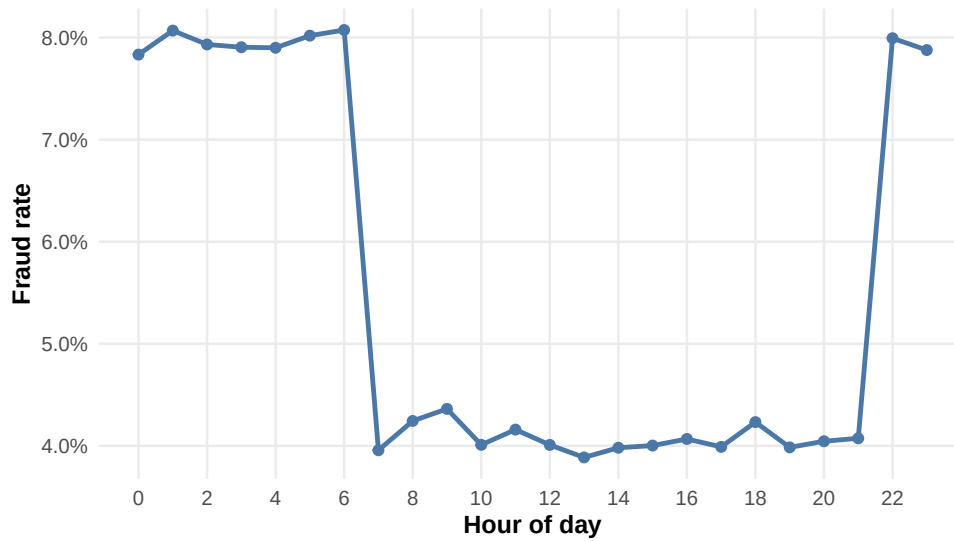


Figure 6: Hourly fraud rate across the 24-hour transaction cycle.

Fraud rates appear higher during late-night and early-morning hours and remain lower through most daytime periods. This agrees with the night-status comparison and shows a strong, repeated timing pattern across adjacent hours. Transaction timing is therefore one of the stronger candidate patterns, although the result is associational rather than causal.

5. Multivariate Fraud Pattern Analysis

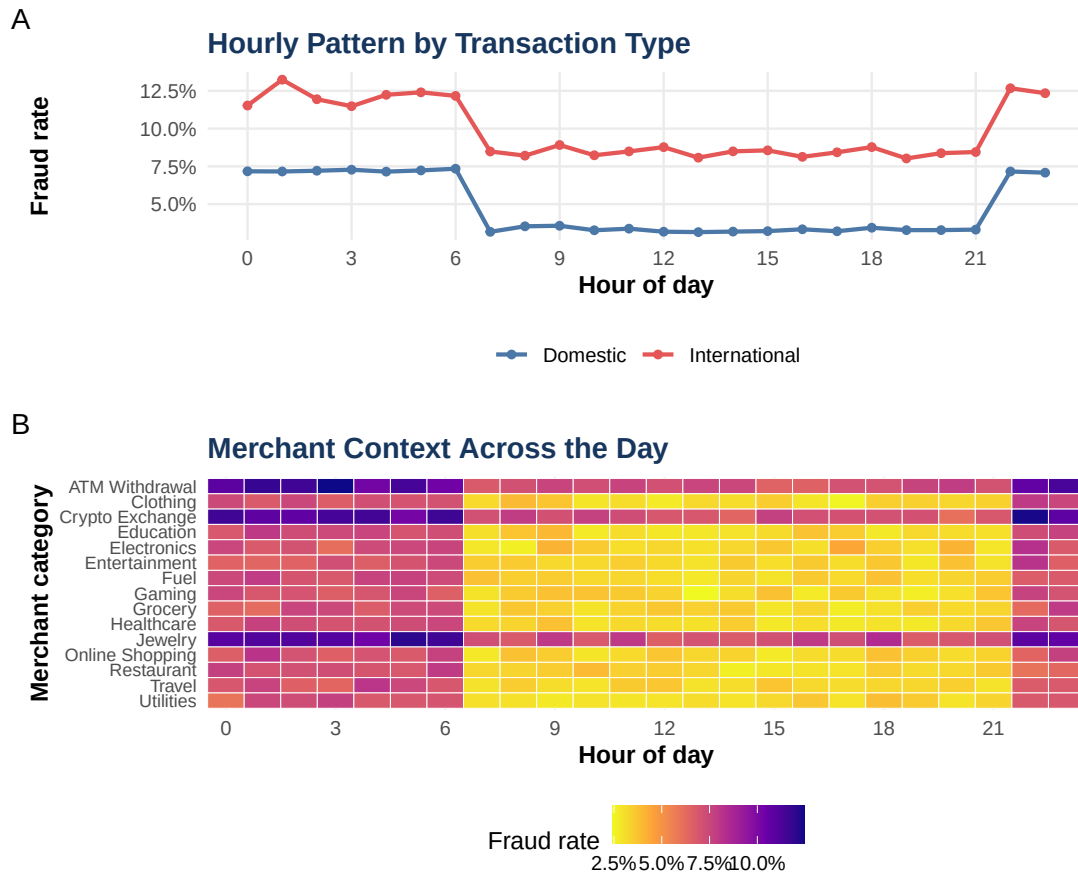


Figure 7: Multivariate fraud patterns: (A) hourly rates by transaction type and (B) merchant-category rates across hours.

Panel A shows that international transactions have higher fraud rates than domestic transactions across most hours. Both groups also show higher fraud rates during late-night and early-morning periods. Panel B adds merchant context and shows that ATM Withdrawal, Jewelry, and Crypto Exchange remain visually higher across multiple hours. Together, these plots suggest that transaction timing and merchant context jointly describe higher fraud-risk patterns in this synthetic dataset.

6. Key Findings and Visual Profile

Fraud is a minority class with an overall rate of 5.53%. The clearest visual patterns involve failed attempts, transaction timing, international status, recent PIN changes, and merchant category. Transactions with two or more failed attempts show substantially higher fraud rates, while late-night and early-morning periods remain elevated across domestic and international groups. ATM withdrawal, jewelry, and crypto exchange transactions appear as the higher-risk merchant contexts. In contrast, transaction amount, payment method, device type, and distance from home show weaker visual separation and are less visually distinctive than the stronger bivariate results.

7. Notes for Figure Selection

Table 3: Figure Shortlist for Team Discussion

Figure name	Analysis type	Recommendation
Binary transaction characteristics	Bivariate	Must use
Fraud rate by merchant category	Bivariate	Must use
Fraud rate by failed attempts	Bivariate	Must use
Fraud rate by hour of day	Bivariate	Must use
Merchant-by-hour heatmap	Multivariate	Must use
Fraud/non-fraud distribution	Dataset overview	Maybe – small overview only
Transaction amount by fraud status	Bivariate	Maybe – cautionary weak finding

The shortlist prioritizes figures that add distinct evidence rather than repeating the same pattern. The combined overview and transaction-amount result remain optional because they provide useful context rather than strong separation.

Limitation. Because the dataset is synthetic, the results should be interpreted as visual patterns within this dataset rather than direct evidence about real-world banking behaviour. Location variables are treated as categorical features, not as verified geographic relationships.